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DeformView: Quantitative Visualization of Non-Linear Deformation Fields

Isabel Frolick [1,2*], **Elise Donszelmann-Lund** [1,2*], **Taj Choksi** [3], **Étienne Léger** [2], **Raphaël Christin** [1,2], **Kaleem Siddiqi** [1], **and D. Louis Collins** [1,2]

1 McGill University, Canada **2** Montreal Neurological Institute, Canada **3** Imperial College London, United Kingdom * These authors contributed equally.

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Summary

DeformView is a new 3D Slicer module (Fedorov et al., 2012) that introduces dense and intuitive voxel-wise maps of displacement magnitude (mm) and local volume change (Jacobian determinant, %) from a transform, and displays them in interactive overlays. Both maps are created as standard Slicer volumes, so they can be saved, exported, and measured with other Slicer tools for quantitative analysis of the deformation. DeformView displays the maps as interactive overlays on the image, with a real-time cursor that reports point-wise deformation values directly on the image and an Increment Transform feature to show show the deformation accumulates over discrete steps. Existing tools in Slicer, such as the Transform module, show only sparse, qualitative representations of the deformation. DeformView is available from our GitHub repository: https://github.com/elisedl1/SlicerDeformView, using the 3D Slicer Extension Wizard.

Statement of need

Non-linear image registration is a key task in medical imaging, where a spatially varying deformation field maps a source image onto a fixed target image. There is an extensive body of work on non-linear registration methods, reflecting its broad applicability across imaging modalities (Ferrante & Paragios, 2017; Sotiras et al., 2013). While the primary output of a registration pipeline is the deformed source image – warped into alignment with the target image – the underlying deformation field itself encodes quantitative information about spatial correspondences and local tissue mechanics.

Clinicians must interpret these resulting deformation fields, either to guide clinical decision-making or to validate that a registration algorithm is performing correctly. However, existing deformation software for non-linear image registration is largely sparse and qualitative, making it difficult to interpret and localize tissue deformation (Brock et al., 2017). For example, how far tissue moved near a structure of interest or where volume was lost or gained. Correct interpretation of deformed anatomy is especially difficult for inexperienced surgeons or researchers.

The displacement magnitude and Jacobian determinant address these concerns bcause they quantify how many millimetres tissue has been displaced near the resection margin.

For example, in image-guided neurosurgery (IGNS), patient-to-image registration is used to align preoperative scans with patient anatomy during surgery. 'Brain shift', or non-linear shifts in brain tissue, causes a mismatch between preoperative images and the patient's current anatomy, necessitating the use of non-linear image registration to deform preoperative images into the intraoperative image space (Abhari et al., 2014). Other use cases include longitudinal

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studies to monitor disease progression or interventional guidance to maximize therapeutic precision. As non-linear image registration is a general problem, these use cases extend across many anatomical systems and modalities.

DeformView addresses these challenges by computing dense, voxel-wise maps of displacement magnitude and local volume change that can be inspected interactively and used in quantitative analysis where deformation is required.

State of the field

The primary existing tool for deformation within 3D Slicer is the Transform module, which offers three sparse representations of the deformation field: glyphs (arrows), a uniform grid, and iso-contours (King et al., 2015). The glyph visualization conveys the direction of local deformation, providing a qualitative sense of how the image has deformed from its initial state.

However, the existing Transform module is purely qualitative with no numerical output, so there is currently no standardized way in Slicer to quantify local deformation magntiude - despite this being a key analytical task in fields like deformable registration. In addition to providing only a sparse, qualitative visualization, existing tools like the Transform module only show the final state deformation; the user is unable to visualize how the deformation is incrementally applied or how specific anatomical regions are compressed or expanded to accommodate the deformation. The absence of these features limits the ability of users to interpret deformation fields.

Software design

DeformView is implemented as a Python extension for 3D Slicer, an open-source platform available on Linux, macOS, and Windows under a BSD-style license. Most Slicer basic infrastructure is implemented in C++ and made available in Python using PythonQt and VTK Python Wrapper (Kapur et al., 2016). The DeformView module is developed using Python following community guidelines for Slicer extension development and built from the official Slicer extension template to ensure consistency and modularity with the Slicer ecosystem (Fedorov et al., 2012).

A key design principle of DeformView is compatibility, both with Slicer's data model and with existing modules. All input selectors expose only the data types relevant to each function: for example, the transformation input only accepts transform types available in Slicer (linear, B-spline, grid, thin-plate spline, and composite) (3D Slicer Community, 2026). As well, DeformView is designed to operate in conjunction with existing Slicer tools. For example, the DeformView colour maps can be displayed alongside the sparse overlays of the existing Transform module. As shown in Figure 1, this combination of the displacement magnitude colour map and glyphs provides both a quantified, spatially localized interpretation and an intuitive representation of local direction change. DeformView includes an automated self-test (built on Slicer's `ScriptedLoadableModuleTest` framework) that verifies the DeformView module without the need for additional downloads or external files, see Validation or the repository README for instructions.

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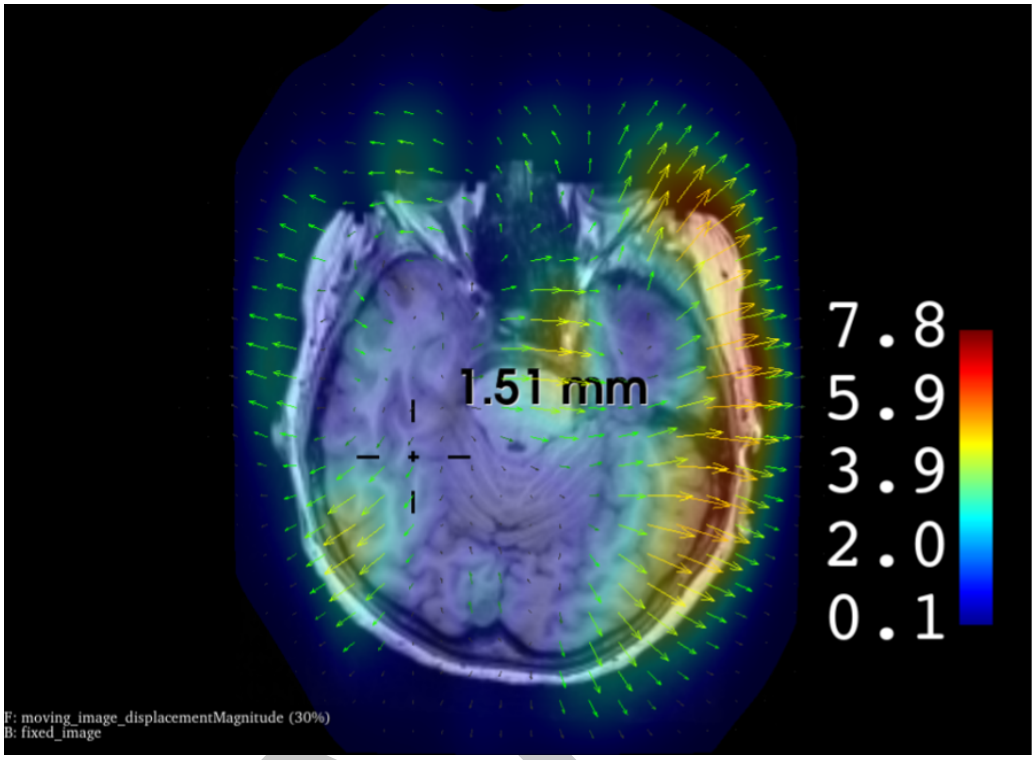


Figure 1: Integration of DeformView displacement magnitude and existing Transform glyph visualization.

Research Impact Statement

DeformView has generated demonstrable interest from the medical imaging research community beyond its core development team. The module was presented at NA-MIC Project Week 43 and 44 (Kapur et al., 2016; NA-MIC, 2025, 2026), the biannual workshop hosted by the National Alliance for Medical Image Computing (NA-MIC), the organization responsible for the continued development of 3D Slicer. DeformView generated discussion among Slicer developers and requests for adoption from external research groups, including for applications in orofacial surgery and mechanical engineering. The DeformView module has also been used in ongoing research on groupwise ultrasound-CT image registration for spinal surgery (Donszelmann-Lund, 2026).

This work was additionally presented as a peer-reviewed poster at the Imaging Network of Ontario (ImNO) 2026 symposium in the Image-Guided Intervention and Surgery category (Donszelmann-Lund et al., 2026).

Overview of DeformView Module

DeformView accepts a deformation field (transform node) and a reference image (typically the deformed source image) as inputs and produces two complementary, dense quantitative visualizations overlaid directly on the image, as shown in Figure 2.

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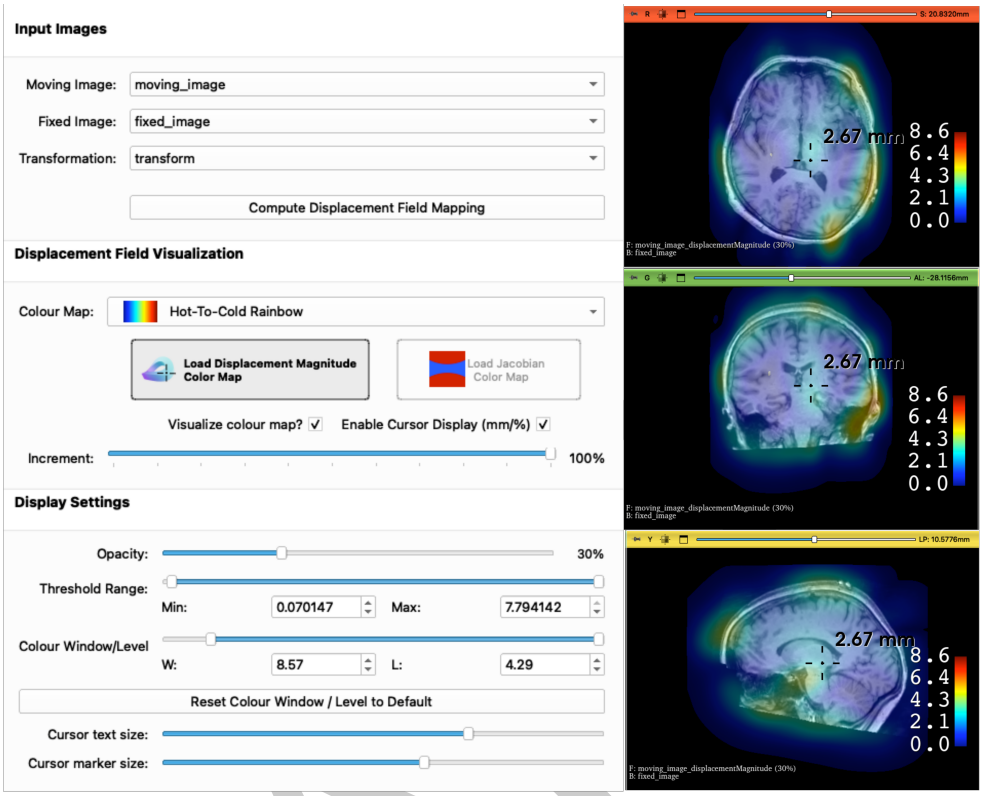


Figure 2: Overview of DeformView 3D Slicer Module. Left: user interface for the proposed module Right: dense colour map (red is higher deformation) with cursor displaying point-wise deformation magnitude on hover in axial (top), coronal (middle), and sagittal (bottom) views

Displacement Magnitude Map

The first new dense colour map renders the Euclidean magnitude of the displacement vector (the distance each voxel has moved between its original and deformed position), in millimeters. Our module converts the transform into a dense vector field sampled across the entire reference image grid, and the length of each displacement vector is stored as a scalar value at each voxel location and displayed using one of eight scientific colour maps, with options for colour-blind readability (Crameri & Hason, 2024). An interactive voxel-wise cursor displays the numerical displacement magnitude at the pointer location in real time, as shown in Figure 3. This provides a dense spatial understanding of deformation quantity and position across the full 3D volume.

Jacobian Determinant Map

This map renders the Jacobian determinant of the deformation field at every voxel as a percentage of local volumetric change, indicating whether a region has expanded or contracted under the deformation (Chung et al., 2001). Tissue expansion (percentage $> 0\%$) is rendered in red, tissue compression (percentage $< 0\%$) in blue, and no change (percentage $= 0\%$) in white, as shown in Figure 3. This allows researchers to identify changes to anatomical regions or areas of physiologically implausible deformation that may indicate registration errors.

Increment Transform

Rather than only displaying the final deformation, as existing modules do, the transformation is incrementally applied to the source image across 10 discrete steps (0.1x, 0.2x, ... 1.0x of the full transform), allowing users to observe how the deformation accumulates spatially. This

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is particularly useful for training and for diagnosing registration behaviour at intermediate stages. It should be noted that, while displacement magnitudes at intermediate steps are exact, intermediate Jacobian values are a linear approximation of the full-transform values.

We visualize the difference between DeformView visualization compared to the existing 3D Slicer functionality in Transform module for both displaying displacement magnitude and local volume change (compression/ expansion) in Figure 3.

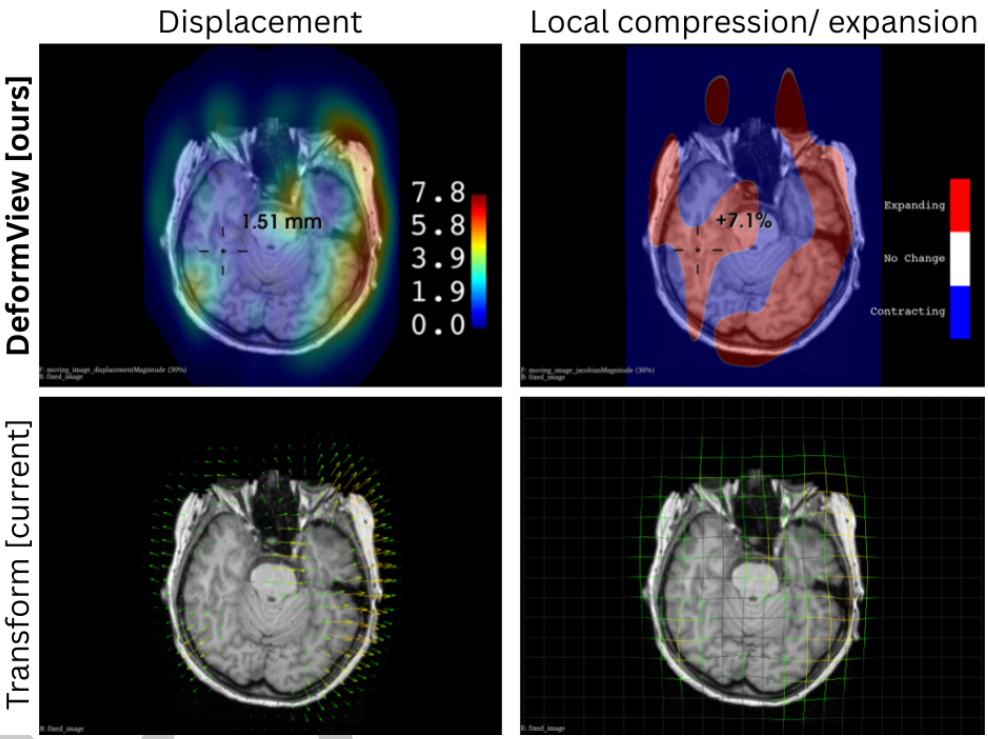


Figure 3: DeformView visualizations (top row) versus existing Transform module visualizations (bottom row). Top left: DeformView displacement magnitude colour map. Top right: DeformView Jacobian colour map. Bottom left: Transform module glyph display. Bottom right: Transform module grid display.

Validation

DeformView includes an automated self-test built on Slicer's `ScriptedLoadableModuleTest` framework that runs within Slicer using only synthetic, in-memory data (see the repository README for instructions). Each test applies a known deformation to a reference volume with anisotropic voxel spacing and compares DeformView's displacement magnitude and Jacobian determinant maps with a reference at every voxel. Translation and affine fields, supplied as linear and grid transforms, and a sinusoidal field supplied as a grid transform are compared against closed-form analytic values. B-spline and thin-plate spline (TPS) transforms, which have no closed-form solution, are compared against Slicer's independent evaluation of the same transform, from which the reference Jacobian is computed by fine finite differences. Results with 3D Slicer 5.8.1 are summarized in Table 1.

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Table 1: Maximum absolute error against the reference. All tests use anisotropic voxel spacing. Jacobian errors are over interior voxels; curved-field values are at 0.5–0.75 mm spacing.

Deformation field	Transform type	Reference	Max. displacement error	Max. Jacobian determinant error
Translation; affine (default and oblique orientation)	Linear	Analytic	$< 10^{-13}$ mm	$< 10^{-14}$
Affine (default and oblique orientation)	Grid	Analytic	$< 10^{-13}$ mm	$< 10^{-14}$
Sinusoidal (non-linear)	Grid	Analytic	$< 10^{-13}$ mm	1.6×10^{-3}
Smooth curved (non-linear)	B-spline	Slicer evaluation	$< 10^{-13}$ mm	1.8×10^{-4}
Smooth curved (non-linear)	Thin-plate spline	Slicer evaluation	$< 10^{-13}$ mm	1.1×10^{-5}

From Table 1, DeformView reproduces every linear deformation field to floating-point precision. On non-linear deformation fields (sinusoidal grid, B-spline, TPS), the Jacobian determinant has a small discretization error because DeformView estimates derivatives by central differences between neighbouring voxels. At every interior voxel, the difference from the reference Jacobian determinant stayed within the theoretical central-difference bound and fell 3.7-4.0x when voxel spacing was halved, as expected for second-order (h^2) accuracy. The largest error at 0.5-0.75 mm spacing was 1.6×10^{-3} on the sinusoidal grid, which was deliberately strongly curved (determinant ranges from 0.44 to 1.89; volume change -56% to +89%) across the field. On less strongly curved non-linear fields, much smaller errors occurred: 1.8×10^{-4} for the B-spline field and 1.1×10^{-5} for the TPS field.

AI usage disclosure

Generative AI tools (Claude Sonnet 4.6) were used in the development of this software, specifically for debugging purposes. Authors have reviewed and validated all AI-assisted code output and made all core design decisions. AI was not used for writing this manuscript, or the preparation of supporting materials.

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